

Generation Interconnection System Impact Study Report

For Queue Project Q15-0499

Roxbury 138 kV

May 20, 2017

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q15-0499. The Interconnection Customer proposes a 142.4 MW battery energy storage facility interconnecting at the Roxbury 138 kV substation in the Pennsylvania area. The request was received on July 11, 2015. The study evaluated system intact and single contingency conditions using the regional planning case, including 204 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 15 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$130,368,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
Milwaukee - Rewey 138 kV	Rewey - Loco Hills 138 kV	172.9	218	25.4
Southbase - Kings 138 kV	Dos Marinas 2003 - Brent 138 kV	159.6	242	8.3
Southbase - Milwaukee 138 kV	Little Chute - Fort Humbug 345 kV	141.4	241	10.5
Lewistown - Loco Hills 138 kV	Edes Tap 2 - W Campus 345 kV	138.7	209	11.6
Rewey - Loco Hills 138 kV	Milwaukee - Rewey Substation 138 kV	138.0	180	18.4
Grand Point - North Lewiston 138 kV line	Rewey - South Bay 1 138 kV line	126.1	182	100.0
Rewey 345/138 kV Autotransformer	Sunrise Ctr - Cayucos 500 kV	124.2	446	30.1
Grand Point - Fayetteville 138 kV Line	Rewey - S Bay 1 138 kV	122.1	173	100.0
Mtn View - Lorich 138 kV Line	Chambersburg - Fair Oaks 345 kV	118.2	223	6.9
Mtn View - Antrim 138 kV	Fair Oaks - Chambersburg 345 kV	116.9	216	6.9
Kings - Greenville Steam 138 kV	Kings - Clayton Substation 138 kV	116.2	190	5.7
Claytn - Kings 138 kV Line	Greenville Steam - Kings 138 kV	112.3	228	6.4
Hazleton - Honk Falls 138 kV	West Campus - Edes Tap 2 345 kV L	108.9	195	5.1
William FMatson Generating Sta 500/345kV TX	N Allegheny Windpower Project - N Yuma 345 kV	107.2	513	21.3
Dpc Argyle - Fayetteville 138 kV L	Rewey Substation - South Bay 1 138 kV	101.4	194	100.0

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Milwaukee - Rewey 138 kV	Rebuild line with higher-capacity conductor	\$4,547,000	100% (\$4,547,000)
Kings - Suthbase 138 kV Line	Rebuild line with higher-capacity conductor	\$8,637,000	100% (\$8,637,000)
Southbase - Milwaukee 138 kV	Rebuild line with higher-capacity conductor	\$11,805,000	100% (\$11,805,000)
Lewistown - Loco Hills 138 kV	Rebuild line with higher-capacity conductor	\$3,660,000	100% (\$3,660,000)
Rewey - Loco Hills 138 kV	Rebuild line with higher-capacity conductor	\$24,369,000	100% (\$24,369,000)
Grand Pt - N Lewiston 138 kV	Rebuild line with higher-capacity conductor	\$1,860,000	100% (\$1,860,000)
Rewey Substation 345/138 kV XFMR	Replace transformer with higher-rated unit	\$16,132,000	100% (\$16,132,000)
Grand Pt Substation - Fayetteville 138 kV	Rebuild line with higher-capacity conductor	\$4,249,000	100% (\$4,249,000)
Mtn View - Lorch 138 kV Line	Rebuild line with higher-capacity conductor	\$2,527,000	18% (\$451,000)
Antrim - Mtn View 138 kV	Rebuild line with higher-capacity conductor	\$5,188,000	18% (\$926,000)
Kings - Greenville Steam 138 kV	Rebuild line with higher-capacity conductor	\$17,786,000	100% (\$17,786,000)
Kings - Clayton 138 kV Line	Rebuild line with higher-capacity conductor	\$13,901,000	100% (\$13,901,000)
Hazleton - Honk Falls 138 kV	Reconductor line	\$1,366,000	100% (\$1,366,000)
William F Matson Generating Sta 500/345 kV Transformer	Replace transformer with higher-rated unit	\$19,494,000	100% (\$19,494,000)
Dpc Argyle - Fayetteville 138 kV	Reconductor line	\$1,185,000	100% (\$1,185,000)
	Total allocated to project		\$130,368,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.